FROM CONSERVATION LAWS TO NAVIER–STOKES

Kinematics, stress, momentum, and boundary data

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right) = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{b}$$

$$\nabla\cdot\boldsymbol{u} = 0$$

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Fluid Dynamics and IBM Theory • Module 0

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GENTLE INTRODUCTIONS TO FLUID DYNAMICS

- **MIT OpenCourseWare: Introduction to Fluid Mechanics (1.060)**
  Conceptual development and physical intuition before detailed analysis.
  https://ocw.mit.edu/courses/1-060-engineering-mechanics-ii-fluid-dynamics-fall-2013/
- **Harvard Natural Sciences Lectures: fluids and biological flows**
  Demonstrations with biological context.
  https://projects.iq.harvard.edu/sciencelectures
- **NPTEL: Fluid Mechanics for Biological Systems**
  A course aimed at life-science audiences.
  https://nptel.ac.in/courses/102103084

ADVANCED UNIVERSITY COURSES

- **MIT OpenCourseWare: 2.25 Advanced Fluid Mechanics**
  https://ocw.mit.edu/courses/2-25-advanced-fluid-mechanics-fall-2013/
- **MIT OpenCourseWare: 2.06 Fluid Dynamics**
  https://ocw.mit.edu/courses/2-06-fluid-dynamics-spring-2020/
- **Cambridge University Engineering Department: Fluid Mechanics I and II notes**
  https://www-civ.eng.cam.ac.uk/cued-lecture-notes/fluids/
- **Caltech GALCIT: fluid-mechanics education resources**
  https://www.galcit.caltech.edu/education

VIDEO LECTURES AND FLOW VISUALIZATION

- **Stanford computational fluid dynamics lectures**
  https://www.youtube.com/playlist?list=PLd3hlSJsX_ImKP68wfKjvRlKq2Ot3PuuX
- **NPTEL: IIT Kanpur Fluid Mechanics**
  https://nptel.ac.in/courses/112104118
- **University of Colorado Boulder: Flow Visualization**
  https://www.flowvis.org/

ADDITIONAL REFERENCES AND DEMONSTRATIONS

- **L. C. Evans: partial differential equations notes**
  https://math.ucdavis.edu/~evans/
- **Dabiri Laboratory: bio-inspired fluid dynamics videos**
  https://dabirilab.com/videos/
- **APS Division of Fluid Dynamics: tutorial videos**
  https://www.aps.org/units/dfd/meetings/video.cfm

ROADMAP

1. **Kinematics:** connect fixed-point measurements to the acceleration of a moving fluid parcel.
2. **Conservation:** derive continuity from mass conservation.
3. **Forces:** represent pressure and viscous surface forces with the Cauchy stress tensor.
4. **Constitutive law:** specialize the momentum balance to an incompressible Newtonian fluid.
5. **Mathematical problem:** pair the equations with compatible initial and boundary data.

Derivation sequence adapted from D. J. Acheson, *Elementary Fluid Dynamics* (1990), Chs. 2 and 6, and G. K. Batchelor, *An Introduction to Fluid Dynamics* (1967), Chs. 2–3. Full references are listed at the end of the deck.

BASIC VARIABLES AND UNITS

Position and velocity

$$\boldsymbol{x} = (x_1, x_2, x_3) = (x, y, z),$$

$$\boldsymbol{u} = (u_1, u_2, u_3) = (u, v, w).$$

Quantity	Symbol	SI unit
density	ρ	$\mathrm{kg\,m^{-3}}$
pressure	p	Pa
dynamic viscosity	μ	$\mathrm{Pa\,s}$
kinematic viscosity	$\nu = \mu/\rho$	$\mathrm{m^2\,s^{-1}}$
body acceleration	$\boldsymbol{b}$	$\mathrm{m\,s^{-2}}$

A body-force density has units of force per volume. Writing it as $\rho\boldsymbol{b}$ distinguishes that density from the acceleration $\boldsymbol{b}$.

Typical values near 20°C: water $\rho \approx 998\,\mathrm{kg\,m^{-3}}$, $\mu \approx 1.0 \times 10^{-3}\,\mathrm{Pa\,s}$, $\nu \approx 1.0 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$; air $\rho \approx 1.2\,\mathrm{kg\,m^{-3}}$, $\mu \approx 1.8 \times 10^{-5}\,\mathrm{Pa\,s}$, $\nu \approx 1.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$.

CARTESIAN VECTOR DIFFERENTIAL OPERATORS

Gradient of a scalar

$$\nabla\phi = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right).$$

Divergence of a vector

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}.$$

Curl of a vector

$$\nabla \times \boldsymbol{u} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}, \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}, \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right).$$

Vector Laplacian

$$\nabla^2\boldsymbol{u} = (\nabla^2 u, \nabla^2 v, \nabla^2 w).$$

SUMMARY OF NOTATION

Symbol	Meaning
$\boldsymbol{x} = (x, y, z)$	spatial position
$\boldsymbol{u} = (u, v, w)$	velocity field
p, ρ, μ	pressure, density, dynamic viscosity
$\nabla\phi$	gradient of a scalar
$\nabla \cdot \boldsymbol{u}$	divergence or local volume-expansion rate
$\nabla \times \boldsymbol{u}$	vorticity vector
$\nabla^2\boldsymbol{u}$	componentwise Laplacian
D/Dt	rate following a fluid parcel

EULERIAN AND MATERIAL DESCRIPTIONS

Eulerian description

The field $\boldsymbol{u}(\boldsymbol{x}, t)$ gives the velocity observed at each fixed spatial point $\boldsymbol{x}$.

This is the natural description for fluid equations on a stationary grid.

Material description

A parcel labeled by $\boldsymbol{\xi}$ follows a trajectory $\mathbf{X}(\boldsymbol{\xi}, t)$ satisfying

$$\frac{d\mathbf{X}}{dt} = \boldsymbol{u}(\mathbf{X}(\boldsymbol{\xi}, t), t).$$

This is the natural description for following matter.

A common choice of label is the initial position, $\boldsymbol{\xi} = \mathbf{X}(\boldsymbol{\xi}, 0)$.

> The material derivative translates an Eulerian field into the rate experienced along a parcel trajectory.

MATERIAL DERIVATIVE OF A SCALAR

Let $\phi(\boldsymbol{x}, t)$ be temperature, concentration, or another scalar field. Along a parcel path,

$$\begin{aligned}\frac{d}{dt}\phi(\mathbf{X}(t), t) &= \frac{\partial \phi}{\partial t} + \frac{d\mathbf{X}}{dt} \cdot \nabla \phi \\ &= \frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi.\end{aligned}$$

Therefore

$$\boxed{\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \boldsymbol{u} \cdot \nabla \phi}.$$

Local change: $\partial \phi / \partial t$ at a fixed point.

Advective change: motion through spatial gradients.

MATERIAL DERIVATIVE IN CARTESIAN COORDINATES

For a scalar field $\phi(x, y, z, t)$ and parcel velocity $\boldsymbol{u} = (u, v, w)$,

$$\frac{d\phi}{dt} = \frac{\partial \phi}{\partial t} + \frac{dx}{dt}\frac{\partial \phi}{\partial x} + \frac{dy}{dt}\frac{\partial \phi}{\partial y} + \frac{dz}{dt}\frac{\partial \phi}{\partial z}.$$

Substituting $dx/dt = u$, $dy/dt = v$, and $dz/dt = w$ gives

$$\boxed{\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} + v\frac{\partial \phi}{\partial y} + w\frac{\partial \phi}{\partial z}}$$

This is the component expansion of $D\phi/Dt = \partial_t \phi + \boldsymbol{u} \cdot \nabla \phi$.

ACCELERATION OF A FLUID PARCEL

Apply the material derivative componentwise to velocity:

$$\boxed{\boldsymbol{a} = \frac{D\boldsymbol{u}}{Dt} = \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}}.$$

In index notation,

$$a_i = \frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j},$$

where repeated indices are summed.

> The advective term is nonlinear because the velocity field transports itself. A steady flow can still accelerate parcels when velocity varies in space.

VECTOR MATERIAL DERIVATIVE IN COMPONENTS

For $\boldsymbol{u} = (u, v, w)$, apply D/Dt to each component:

$$\frac{Du}{Dt} = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z},$$

$$\frac{Dv}{Dt} = \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z},$$

$$\frac{Dw}{Dt} = \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}.$$

$$\frac{D\boldsymbol{u}}{Dt} = \left(\frac{Du}{Dt}, \frac{Dv}{Dt}, \frac{Dw}{Dt}\right).$$

LOCAL AND ADVECTIVE ACCELERATION

$$\frac{D\boldsymbol{u}}{Dt} = \underbrace{\frac{\partial \boldsymbol{u}}{\partial t}}_{\text{local acceleration}} + \underbrace{(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}}_{\text{advective acceleration}} .$$

The local term measures change at a fixed spatial point.

The advective term measures change as a parcel moves through a nonuniform velocity field.

> A flow can be steady, $\partial_t \boldsymbol{u} = 0$, while individual parcels accelerate. Flow through a narrowing passage is a familiar example.

PHYSICAL MEANING OF INCOMPRESSIBILITY

Imagine a small dyed blob that contains the same fluid parcels as it moves.

The blob may translate, rotate, stretch, and shear. For an incompressible material, however, its volume remains constant:

$$\boxed{\frac{d}{dt}|\Omega(t)| = 0}.$$

> Incompressibility restricts local volume change. It does not require the blob to retain its shape, and it does not mean that the velocity is spatially uniform.

The next step expresses the blob's volume change in terms of flux through its moving boundary.

VOLUME FLUX AND THE DIVERGENCE THEOREM

Let $\Omega(t)$ be a material region with outward unit normal $\boldsymbol{n}$. Its boundary moves with the fluid, so

$$\frac{d}{dt}|\Omega(t)| = \oint_{\partial\Omega(t)} \boldsymbol{u}\cdot\boldsymbol{n}\, dS.$$

The divergence theorem converts this surface flux into a volume integral:

$$\oint_{\partial\Omega(t)} \boldsymbol{u}\cdot\boldsymbol{n}\, dS = \int_{\Omega(t)} \nabla\cdot\boldsymbol{u}\, dV.$$

If every material region preserves volume, then

$$\boxed{\nabla\cdot\boldsymbol{u} = 0}.$$

> Positive divergence means local expansion; negative divergence means local compression.

The first identity is the Reynolds transport theorem applied to the integrand 1; the step to $\nabla\cdot\boldsymbol{u}=0$ uses that the region is arbitrary and $\nabla\cdot\boldsymbol{u}$ is continuous.

MASS CONSERVATION FOR A MATERIAL REGION

Let $\Omega(t)$ contain the same fluid parcels for all time. Its mass is

$$M(t) = \int_{\Omega(t)} \rho(\boldsymbol{x}, t)\, dV.$$

Mass conservation requires

$$\frac{d}{dt} \int_{\Omega(t)} \rho\, dV = 0.$$

The Reynolds transport theorem gives

$$\frac{d}{dt} \int_{\Omega(t)} \rho\, dV = \int_{\Omega(t)} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) \right] dV.$$

Because the material region is arbitrary (and the integrand is continuous), the integrand must vanish.

Reynolds transport theorem: for a smooth field F and material region $\Omega(t)$, $\frac{d}{dt} \int_{\Omega(t)} F\, dV = \int_{\Omega(t)} \left[\partial_t F + \nabla \cdot (F \boldsymbol{u}) \right] dV$.

CONTINUITY AND INCOMPRESSIBILITY

The differential mass balance is

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0} \qquad \Longleftrightarrow \qquad \boxed{\frac{D\rho}{Dt} + \rho \nabla \cdot \boldsymbol{u} = 0}.$$

For an incompressible material, parcel density does not change:

$$\frac{D\rho}{Dt} = 0 \qquad \Longrightarrow \qquad \boxed{\nabla \cdot \boldsymbol{u} = 0}.$$

The volume of a material region satisfies

$$\frac{d}{dt}|\Omega(t)| = \oint_{\partial\Omega(t)} \boldsymbol{u} \cdot \boldsymbol{n}\, dS = \int_{\Omega(t)} \nabla \cdot \boldsymbol{u}\, dV.$$

Uniform constant density is a stronger assumption than incompressibility: a divergence-free flow may advect spatial density variations.

PRESSURE TRACTION ON A FLUID PARCEL

Let $\Omega(t)$ be a small material region with outward unit normal $\boldsymbol{n}$.

Pressure is an isotropic compressive stress: its magnitude is independent of the orientation of the surface. The traction exerted by the surrounding fluid on the parcel is

$$\boxed{\boldsymbol{t}_p(\boldsymbol{n}) = -p\boldsymbol{n}}.$$

Thus the force on a surface element dS is $-p\boldsymbol{n}\,dS$.

> Because $\boldsymbol{n}$ points out of the parcel, the minus sign makes a positive pressure act inward.

NET PRESSURE FORCE

Adding the pressure traction over the parcel boundary gives

$$\boxed{\mathbf{F}_p = -\oint_{\partial\Omega(t)} p\boldsymbol{n}\, dS}.$$

For constant pressure, contributions on opposite sides cancel and the net force is zero.

For spatially varying pressure, the higher-pressure side pushes more strongly. The resultant points toward lower pressure.

> The opposite force, exerted by the parcel on its surroundings across the same interface, follows from Newton's third law.

FROM SURFACE PRESSURE TO A VOLUME FORCE

Apply the divergence theorem componentwise. For each Cartesian basis vector $\boldsymbol{e}_i$,

$$\oint_{\partial\Omega} (p\boldsymbol{e}_i) \cdot \boldsymbol{n}\, dS = \int_{\Omega} \nabla \cdot (p\boldsymbol{e}_i)\, dV = \int_{\Omega} \frac{\partial p}{\partial x_i}\, dV.$$

Stacking the components gives

$$\oint_{\partial\Omega} p\boldsymbol{n}\, dS = \int_{\Omega} \nabla p\, dV.$$

Therefore

$$\boxed{\mathbf{F}_p = -\int_{\Omega} \nabla p\, dV}.$$

SMALL-PARCEL LIMIT AND SIGN CHECK

If ∇p is nearly constant within a parcel of volume δV,

$$\boxed{\mathbf{F}_{p,\mathrm{parcel}} \approx -(\nabla p)\,\delta V}.$$

The force per unit volume is therefore $-\nabla p$.

For the same interface and orientation, Newton's third law gives the opposite resultant exerted by the parcel on the surrounding fluid:

$$\mathbf{F}_{p,\mathrm{surroundings}} \approx +(\nabla p)\,\delta V.$$

> Always state whose force is being calculated and which region defines the outward normal.

IDEAL-FLUID SPECIALIZATION USED HERE

The introductory Euler model combines three assumptions:

- **inviscid stress**: $\boldsymbol{\sigma} = -p\mathbf{I}$;
- **incompressible motion**: $\nabla \cdot \boldsymbol{u} = 0$;
- **uniform constant density**: ρ is the same for every parcel and time.

Pressure supplies the surface traction

$$\boldsymbol{t}_p = -p\boldsymbol{n}.$$

> Inviscid, incompressible, and constant-density are distinct assumptions. Compressible Euler equations retain the inviscid stress model while allowing density to vary.

MOMENTUM OF A DYED PARCEL: APPLIED FORCES

For a small material parcel of volume δV,

$$\mathbf{F}_p = -\oint_{\partial\Omega} p\boldsymbol{n}\, dS = -\int_{\delta V} \nabla p\, dV \approx -(\nabla p)\, \delta V.$$

If gravity is the body acceleration, then

$$\mathbf{F}_g = \int_{\delta V} \rho\boldsymbol{g}\, dV \approx \rho\boldsymbol{g}\, \delta V.$$

Hence

$$\boxed{\mathbf{F}_{\text{total}} \approx (-\nabla p + \rho\boldsymbol{g})\, \delta V}.$$

MOMENTUM OF A DYED PARCEL: NEWTON'S LAW

The parcel mass is $\rho\,\delta V$, and its acceleration is the material derivative $D\boldsymbol{u}/Dt$. Newton's second law gives

$$\rho\,\delta V\,\frac{D\boldsymbol{u}}{Dt} = (-\nabla p + \rho\boldsymbol{g})\,\delta V.$$

Canceling δV yields

$$\boxed{\rho\frac{D\boldsymbol{u}}{Dt} = -\nabla p + \rho\boldsymbol{g}}.$$

> The same result follows exactly from an integral momentum balance for an arbitrary material region; the small-parcel argument supplies the physical picture.

EULER EQUATIONS

Using $D/Dt = \partial_t + \boldsymbol{u} \cdot \nabla$, the inviscid momentum balance is

$$\boxed{\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\frac{1}{\rho}\nabla p + \boldsymbol{b}}.$$

From here the general body acceleration $\boldsymbol{b}$ replaces the gravitational acceleration used in the parcel derivation (with $\boldsymbol{b} = \boldsymbol{g}$ for gravity).
For the constant-density incompressible specialization,

$$\boxed{\nabla \cdot \boldsymbol{u} = 0}.$$

> Pressure adjusts to enforce the divergence constraint while the momentum equation determines parcel acceleration.

EULER EQUATIONS IN CARTESIAN COMPONENTS

For $\boldsymbol{u} = (u, v, w)$ and $\boldsymbol{b} = (b_x, b_y, b_z)$,

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + b_x,$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + b_y,$$

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + b_z.$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

THE STRESS VECTOR

At a point $\boldsymbol{x}$, consider a small oriented surface element with unit normal $\boldsymbol{n}$. The force transmitted across it is

$$\boldsymbol{t}(\boldsymbol{n})\, dS,$$

where $\boldsymbol{t}$ is the traction or stress vector.

- In an inviscid fluid, $\boldsymbol{t} = -p\boldsymbol{n}$.
- In a viscous fluid, $\boldsymbol{t}$ may contain normal and tangential components.
- Reversing the normal reverses the traction: $\boldsymbol{t}(-\boldsymbol{n}) = -\boldsymbol{t}(\boldsymbol{n})$.

> Cauchy's theorem expresses the traction on every possible plane through one tensor field defined at the point.

COMPONENTS OF THE CAUCHY STRESS

Define

$$\sigma_{ij} = \text{the } i\text{th component of force per unit area on a plane whose normal is } \boldsymbol{e}_j.$$

In Cartesian coordinates,

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix}.$$

Each column corresponds to a coordinate-plane normal; each row gives a force component.

The tensor must determine the traction on an arbitrarily oriented plane.

STRESS ON AN ARBITRARY SURFACE

Applying linear momentum balance to a shrinking tetrahedron gives Cauchy's stress principle:

$$\boxed{\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{\sigma}\boldsymbol{n}}, \qquad t_i = \sigma_{ij} n_j.$$

Surface forces scale with length squared while inertia and body forces scale with length cubed. The surface-force balance therefore determines the limiting relation.

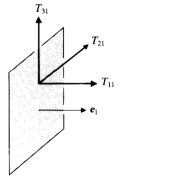


Fig. 6.2. Three components of the stress tensor T_{ij}.

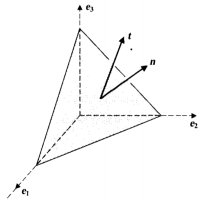


Fig. 6.3. Definition sketch for the proof of eqn (6.3).

Stress components and the Cauchy tetrahedron. Acheson writes the stress tensor T_{ij} and the traction $\boldsymbol{t}$; we write σ_{ij} and $\boldsymbol{t}$. Source: D. J. Acheson, *Elementary Fluid Dynamics* (Oxford Univ. Press, 1990), Figs. 6.2–6.3.

LINEAR MOMENTUM BALANCE WITH STRESSES

For a material region $\Omega(t)$,

$$\frac{d}{dt}\int_{\Omega(t)} \rho \boldsymbol{u}\, dV = \underbrace{\oint_{\partial\Omega(t)} \boldsymbol{\sigma}\boldsymbol{n}\, dS}_{\text{surface force}} + \underbrace{\int_{\Omega(t)} \rho \boldsymbol{b}\, dV}_{\text{body force}}.$$

The divergence theorem gives

$$\oint_{\partial\Omega(t)} \boldsymbol{\sigma}\boldsymbol{n}\, dS = \int_{\Omega(t)} \nabla\cdot\boldsymbol{\sigma}\, dV.$$

Using mass conservation (Reynolds transport theorem plus continuity),

$$\frac{d}{dt}\int_{\Omega(t)} \rho \boldsymbol{u}\, dV = \int_{\Omega(t)} \rho \frac{D\boldsymbol{u}}{Dt}\, dV,$$

because the mass $\rho\, dV$ of each parcel is constant in time.

so that

$$\int_{\Omega(t)} \rho \frac{D\boldsymbol{u}}{Dt}\, dV = \int_{\Omega(t)} (\nabla\cdot\boldsymbol{\sigma} + \rho\boldsymbol{b})\, dV.$$

CAUCHY MOMENTUM EQUATION

Because the material region is arbitrary,

$$\boxed{\rho \frac{D\boldsymbol{u}}{Dt} = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{b}}.$$

For an inviscid fluid, $\boldsymbol{\sigma} = -p\mathbf{I}$, so

$$\nabla \cdot \boldsymbol{\sigma} = -\nabla p$$

and Cauchy's equation reduces to the Euler momentum equation.

> This is Newton's second law per unit volume. The balance law is general; a constitutive relation for $\boldsymbol{\sigma}$ specifies the fluid model.

TRACTION AND MOMENTUM IN INDEX NOTATION

The traction relation and momentum equation use the same divergence convention:

$$t_i = \sigma_{ij} n_j, \qquad (\nabla \cdot \boldsymbol{\sigma})_i = \frac{\partial \sigma_{ij}}{\partial x_j}.$$

Hence

$$\boxed{\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = \frac{\partial \sigma_{ij}}{\partial x_j} + \rho b_i}.$$

> The first stress index gives force direction; the second identifies the surface-normal direction.

PRESSURE AND VISCOUS PARTS OF STRESS

Split the Cauchy stress as

$$\boxed{\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}},$$

where $-p\mathbf{I}$ is isotropic and $\boldsymbol{\tau}$ is the viscous or extra stress. Then

$$\rho \frac{D\boldsymbol{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \boldsymbol{b}.$$

For a classical fluid without couple stresses, angular-momentum balance gives

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathsf{T}}.$$

> A constitutive law for $\boldsymbol{\tau}$ closes the momentum equation.

DEFORMATION AND ROTATION RATES

Decompose the velocity gradient into symmetric and antisymmetric parts:

$$\nabla \boldsymbol{u} = \mathbf{D} + \mathbf{W},$$

$$\mathbf{D} = \frac{1}{2}\left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^{\mathsf{T}}\right), \qquad \mathbf{W} = \frac{1}{2}\left(\nabla \boldsymbol{u} - (\nabla \boldsymbol{u})^{\mathsf{T}}\right).$$

$\mathbf{D}$: rate of stretching and shearing,

$$\operatorname{tr} \mathbf{D} = \nabla \cdot \boldsymbol{u}.$$

$\mathbf{W}$: local rigid-body rotation rate.
For an isotropic Newtonian simple fluid, $\mathbf{W}$ does not contribute to viscous stress.

GEOMETRIC INTERPRETATION OF THE VELOCITY GRADIENT

For two nearby parcels separated by $d\boldsymbol{x}$, their relative velocity is

$$d\boldsymbol{u} = (\nabla \boldsymbol{u})\, d\boldsymbol{x} = \mathbf{D}\, d\boldsymbol{x} + \mathbf{W}\, d\boldsymbol{x}.$$

Symmetric part $\mathbf{D}$

- changes lengths;
- changes angles;
- measures stretching and shearing.

Antisymmetric part $\mathbf{W}$

- rotates line elements;
- preserves lengths and angles;
- represents local rigid-body spin.

> An objective viscous constitutive law cannot generate stress from a superposed rigid-body rotation alone.

REQUIREMENTS FOR A NEWTONIAN CONSTITUTIVE LAW

The extra stress $\boldsymbol{\tau}$ for a Newtonian simple fluid is modeled as:

1. **linear** in the local deformation rate;
2. **local** in space and time;
3. **objective**, so rigid-body motion alone produces no viscous stress;
4. **isotropic**, so the material has no preferred direction.

For an isotropic material, the available linear tensor ingredients are $\mathbf{D}$, its trace $\nabla \cdot \boldsymbol{u}$, and $\mathbf{I}$. Therefore

$$\boldsymbol{\tau} = 2\mu\mathbf{D} + \lambda(\nabla \cdot \boldsymbol{u})\mathbf{I}.$$

NEWTONIAN CONSTITUTIVE LAW

For a linear, local, isotropic Newtonian fluid,

$$\boxed{\boldsymbol{\tau} = 2\mu\mathbf{D} + \lambda(\nabla \cdot \boldsymbol{u})\mathbf{I}}.$$

- μ is the dynamic shear viscosity.
- λ is the second viscosity coefficient.
- In three dimensions, the bulk viscosity is

$$\zeta = \lambda + \frac{2}{3}\mu.$$

- Non-negative dissipation requires $\mu \geq 0$ and $\zeta \geq 0$; Stokes' hypothesis $\zeta = 0$ gives $\lambda = -\frac{2}{3}\mu$.

> Isotropy means the response has no fixed material direction and transforms consistently under rotations.

INCOMPRESSIBLE NEWTONIAN STRESS

When $\nabla \cdot \boldsymbol{u} = 0$,

$$\boxed{\boldsymbol{\tau} = 2\mu \mathbf{D}}, \qquad \boxed{\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu \mathbf{D}}.$$

In components,

$$\sigma_{ij} = -p\delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right).$$

Pressure supplies isotropic normal stress.

Viscosity supplies shear and extensional normal stresses proportional to deformation rate.

VISCOUS FORCE DENSITY

For constant viscosity,

$$\nabla \cdot (2\mu \mathbf{D}) = \mu \nabla^2 \boldsymbol{u} + \mu \nabla (\nabla \cdot \boldsymbol{u}).$$

In index form: $\partial_j[\mu(\partial_j u_i + \partial_i u_j)] = \mu\, \partial_j \partial_j u_i + \mu\, \partial_i (\partial_j u_j)$, since mixed partial derivatives commute. Therefore, for incompressible flow,

$$\boxed{\nabla \cdot \boldsymbol{\tau} = \mu \nabla^2 \boldsymbol{u}}.$$

The Laplacian acts componentwise and diffuses momentum from regions of high velocity toward regions of low velocity.

> If μ varies in space, keep $\nabla \cdot (2\mu \mathbf{D})$ in conservative form; gradients of viscosity generate additional terms.

INCOMPRESSIBLE NAVIER–STOKES EQUATIONS

Combining momentum balance with the incompressible Newtonian constitutive law gives

$$\boxed{\rho\left[\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u}\right] = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{b}}$$

$$\boxed{\nabla\cdot\boldsymbol{u} = 0}.$$

Equivalently, using kinematic viscosity $\nu = \mu/\rho$,

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{b}.$$

NAVIER–STOKES IN CARTESIAN PARTIAL DERIVATIVES

For $\boldsymbol{u} = (u, v, w)$, constant ρ and μ, and $\boldsymbol{b} = (b_x, b_y, b_z)$,

$$\begin{aligned}
\rho\left(u_t + uu_x + vu_y + wu_z\right) &= -p_x + \mu(u_{xx} + u_{yy} + u_{zz}) + \rho b_x, \\
\rho\left(v_t + uv_x + vv_y + wv_z\right) &= -p_y + \mu(v_{xx} + v_{yy} + v_{zz}) + \rho b_y, \\
\rho\left(w_t + uw_x + vw_y + ww_z\right) &= -p_z + \mu(w_{xx} + w_{yy} + w_{zz}) + \rho b_z.
\end{aligned}$$

The incompressibility constraint is

$$u_x + v_y + w_z = 0.$$

> These are three momentum equations coupled through pressure, incompressibility, and the advective terms.

TERMS IN THE MOMENTUM EQUATION

Every term below has units of force per volume, $\mathrm{N\,m^{-3}}$.

Term	Physical role
$\rho\dfrac{\partial \boldsymbol{u}}{\partial t}$	local inertia
$\rho(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$	advective inertia
$-\nabla p$	pressure-gradient force
$\mu\nabla^2\boldsymbol{u}$	viscous momentum diffusion
$\rho\boldsymbol{b}$	body force

> The Reynolds number compares the advective-inertia scale $\rho U^2/L$ with the viscous scale $\mu U/L^2$.

CARTESIAN INDEX FORM

Using Einstein summation (every repeated spatial index is summed from 1 to 3),

$$\boxed{\rho\left(\frac{\partial u_i}{\partial t}+u_j\frac{\partial u_i}{\partial x_j}\right)=-\frac{\partial p}{\partial x_i}+\mu\frac{\partial^2 u_i}{\partial x_j\partial x_j}+\rho b_i}, \qquad i=1,2,3,$$

with

$$\boxed{\frac{\partial u_i}{\partial x_i}=0}.$$

> Index notation keeps the vector structure visible without placing three long component equations on one slide.

STEADY INCOMPRESSIBLE NAVIER–STOKES EQUATIONS

For a steady flow, $\partial \boldsymbol{u}/\partial t = 0$. The equations become

$$\boxed{\rho(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{b}}, \qquad \boxed{\nabla\cdot\boldsymbol{u} = 0}.$$

- Parcel acceleration may remain nonzero through advection.
- The equations remain nonlinear unless the assumed geometry eliminates $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$.
- Fully developed channel and pipe flows provide useful analytical examples.

FULLY DEVELOPED UNIDIRECTIONAL FLOW

Assume a parallel flow

$$\boldsymbol{u} = (u(y,t), 0, 0), \qquad \frac{\partial u}{\partial x} = \frac{\partial u}{\partial z} = 0.$$

Incompressibility is automatic, and the advective term vanishes. The streamwise equation reduces to

$$\boxed{\rho \frac{\partial u}{\partial t} = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} + \rho b_x}.$$

Couette flow: wall motion drives $u(y)$.

Poiseuille flow: a pressure gradient drives $u(y)$.

The y- and z-equations reduce to $\partial p/\partial y = \rho b_y$ and $\partial p/\partial z = \rho b_z$. For uniform $\boldsymbol{b}$, $\partial p/\partial x$ can then depend only on t, so the streamwise pressure gradient is uniform in space.

TWO-DIMENSIONAL INCOMPRESSIBLE FLOW

Let $\boldsymbol{u} = (u(x, y, t), v(x, y, t), 0)$. Then

$$\rho(u_t + uu_x + vu_y) = -p_x + \mu(u_{xx} + u_{yy}) + \rho b_x,$$

$$\rho(v_t + uv_x + vv_y) = -p_y + \mu(v_{xx} + v_{yy}) + \rho b_y,$$

with

$$\boxed{u_x + v_y = 0}.$$

> Two-dimensional models support nonlinear advection, separation, and vortex shedding, but omit three-dimensional instabilities and vortex stretching.

BOUNDARY CONDITIONS FOR NAVIER–STOKES

The equations must be paired with conditions on $\partial\Omega$ that represent the physical boundary.

- A solid wall constrains fluid motion relative to the wall.
- An inflow specifies incoming velocity or flux.
- An open boundary controls traction or another outflow model.
- An interface couples kinematics to a traction balance.
- Periodic faces identify values across repeating boundaries.

> Boundary conditions are part of the physical and mathematical model. Changing them can change the solution even when the differential equations remain the same.

INITIAL AND BOUNDARY DATA

An unsteady incompressible problem needs an initial velocity

$$\boldsymbol{u}(\boldsymbol{x}, 0) = \boldsymbol{u}_0(\boldsymbol{x}), \qquad \nabla \cdot \boldsymbol{u}_0 = 0,$$

compatible with the boundary data at $t = 0$.
Partition the boundary into portions with velocity and traction data:

$$\partial\Omega = \Gamma_D \cup \Gamma_N, \qquad \Gamma_D \cap \Gamma_N = \varnothing,$$

$$\boldsymbol{u} = \overline{\boldsymbol{u}} \text{ on } \Gamma_D, \qquad \boldsymbol{\sigma n} = \bar{\boldsymbol{t}} \text{ on } \Gamma_N.$$

For velocity prescribed on the entire boundary, incompressibility requires

$$\int_{\partial\Omega} \overline{\boldsymbol{u}} \cdot \boldsymbol{n} \, dS = 0.$$

SOLID-WALL AND INFLOW CONDITIONS

No-slip solid wall

$$\boldsymbol{u} = \mathbf{U}_{\text{wall}}.$$

For a stationary wall, $\boldsymbol{u} = \mathbf{0}$.

No penetration only

$$(\boldsymbol{u} - \mathbf{U}_{\text{wall}}) \cdot \boldsymbol{n} = 0.$$

This constrains the normal velocity component but still requires a tangential condition.

Prescribed inflow

$$\boldsymbol{u} = \boldsymbol{u}_{\text{in}}(\boldsymbol{x}, t).$$

The profile should be physically plausible and compatible with the net outflow.

> Dirichlet data specify velocity. No penetration is an essential condition on one component, not a traction condition.

TRACTION AND OPEN-BOUNDARY CONDITIONS

A traction condition prescribes force per unit area:

$$\boldsymbol{\sigma n} = \bar{\boldsymbol{t}}.$$

Examples include:

- stress-free idealization: $\boldsymbol{\sigma n} = \mathbf{0}$;
- known external pressure with negligible external shear:

$$\boldsymbol{\sigma n} = -p_{\mathrm{ext}}\boldsymbol{n};$$

- a variational “do-nothing” outflow, whose precise traction depends on the chosen weak form.

On do-nothing and other artificial outflow conditions, see Heywood, Rannacher, and Turek (1996).

> A zero normal velocity gradient is not a universal outflow condition. Open boundaries require a condition consistent with the flow direction, stress model, and numerical formulation.

ROBIN AND NAVIER-SLIP CONDITIONS

A Robin condition combines a velocity response with a traction response. A common example is Navier slip:

$$(\boldsymbol{u} - \mathbf{U}_{\text{wall}}) \cdot \boldsymbol{n} = 0,$$

$$(\boldsymbol{\sigma n})_{\parallel} + \frac{\mu}{\ell_s}(\boldsymbol{u} - \mathbf{U}_{\text{wall}})_{\parallel} = 0.$$

Here ℓ_s is the slip length and $\boldsymbol{n}$ is the outward normal of the fluid domain.

- $\ell_s \to 0$ approaches no slip.
- Large ℓ_s approaches small tangential traction.
- Resistive or porous boundaries use related mixed relations.

SLIP, PERIODIC, AND FREE-SURFACE CONDITIONS

Navier slip

$$(\boldsymbol{u} - \mathbf{U}_{\text{wall}}) \cdot \boldsymbol{n} = 0,$$

$$(\boldsymbol{\sigma n})_{\parallel} + \frac{\mu}{\ell_s}(\boldsymbol{u} - \mathbf{U}_{\text{wall}})_{\parallel} = 0.$$

The displayed sign uses the outward normal of the fluid domain. (Recap of the previous slide, repeated beside the other conditions for comparison. For a stationary wall at $y = 0$ below the fluid, $\boldsymbol{n} = -\boldsymbol{e}_y$ and the condition reduces to $u = \ell_s \, \partial u/\partial y$.)

Periodic flow

Velocity is periodic. Pressure may be periodic up to a prescribed pressure drop.

Free surface

Two conditions are needed:

- a kinematic condition that moves the interface with the fluid;
- a dynamic condition that balances the traction jump with surface-tension and Marangoni forces.

> State the normal and curvature conventions when writing the surface-tension sign.

CHOOSING BOUNDARY CONDITIONS

Match each condition to the physics and to the mathematical formulation:

- use velocity data where motion is known, such as an inflow or no-slip wall;
- use traction or a formulation-specific open condition where force exchange is known;
- use periodic data only when the geometry and forcing repeat;
- state both kinematic and dynamic conditions at a free surface;
- test outflow placement and sensitivity because artificial open boundaries are delicate.

> Do not prescribe more information than the coupled velocity-pressure system can accept. Check flux compatibility and the pressure gauge.

PRESSURE LEVEL AND PRESSURE BOUNDARY DATA

In the momentum equation, pressure appears through ∇p.

- With velocity prescribed everywhere, p is determined only up to an additive constant. Fix a gauge such as

$$\int_\Omega p \, dV = 0 \qquad \text{or} \qquad p(\boldsymbol{x}_0) = 0.$$

- A traction boundary can determine the pressure level because p is part of $\boldsymbol{\sigma n}$.
- A “pressure outlet” is normally shorthand for a traction model; viscous and tangential stresses must be handled consistently.

> Prescribing a complete velocity and a complete traction on the same boundary segment generally overdetermines the standard incompressible formulation.

PRESSURE POISSON EQUATION

Take the divergence of the constant-density, constant-viscosity momentum equation and use $\nabla \cdot \boldsymbol{u} = 0$:

$$\boxed{\nabla^2 p = -\rho \nabla \cdot [(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}] + \rho \nabla \cdot \boldsymbol{b}}.$$

For constant gravity, $\nabla \cdot \boldsymbol{b} = 0$. Because $\nabla \cdot \boldsymbol{u} = 0$, the advective source can be written $\nabla \cdot [(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}] = \dfrac{\partial u_i}{\partial x_j}\dfrac{\partial u_j}{\partial x_i}$.

The pressure equation inherits boundary information from the normal momentum balance and the chosen velocity or traction conditions.

> Pressure acts as a Lagrange multiplier enforcing incompressibility. The Poisson equation is a consequence of momentum balance, not an independent constitutive law.

BOUNDARY-DATA CHECKLIST

1. Match the condition to the physical boundary: wall, inflow, outflow, interface, or periodic face.
2. Avoid prescribing full velocity and full traction on the same boundary segment.
3. Enforce incompressible flux compatibility for prescribed velocity data.
4. Supply divergence-free, boundary-compatible initial data for unsteady flow.
5. Fix the pressure gauge only when the boundary conditions leave it undetermined.

> Boundary conditions are part of the mathematical model; small changes can produce different physical solutions.

WELL-POSED AND OVERDETERMINED EXAMPLES

Compatible examples

- prescribed inflow velocity with a consistent traction condition at outflow;
- no-slip walls with traction specified on the remaining open boundary;
- periodic velocity with a pressure gauge or a prescribed mean pressure drop.

Common problems

- prescribing complete velocity and complete traction on the same boundary segment;
- prescribing incompatible net inflow when velocity is fixed everywhere;
- treating a scalar pressure value as a complete open-boundary condition without specifying the accompanying viscous traction model.

> Well-posedness also depends on domain geometry, regularity, and the exact weak or numerical formulation; these examples are modeling guides, not a general theorem.

ASSUMPTIONS AND EXTENSIONS

The derived Navier–Stokes form assumes:

- a continuum fluid;
- incompressibility and uniform constant density;
- a Newtonian, isotropic constitutive law;
- constant dynamic viscosity;
- classical Cauchy stress without couple stresses.

Extensions modify specific parts of the derivation:

- compressible flow couples momentum to density and energy;
- variable or non-Newtonian viscosity changes $\nabla \cdot \boldsymbol{\tau}$;
- moving interfaces add kinematic and traction-jump conditions.

DERIVATION SUMMARY

1. Parcel acceleration is $D\boldsymbol{u}/Dt = \partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}$.
2. Mass conservation gives continuity; incompressibility gives $\nabla \cdot \boldsymbol{u} = 0$.
3. Surface force is represented by traction $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$.
4. Momentum balance gives $\rho D\boldsymbol{u}/Dt = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{b}$.
5. The incompressible Newtonian law $\boldsymbol{\sigma} = -p\mathbf{I} + 2\mu\mathbf{D}$ yields Navier–Stokes.

> Conservation laws determine the balances. The constitutive law and boundary data determine the particular fluid model and problem.

FURTHER STUDY

- MIT OpenCourseWare 1.060: Engineering Mechanics II for an introductory treatment.
- MIT OpenCourseWare 2.25: Advanced Fluid Mechanics for graduate-level notes and problems.
- Harvard Natural Sciences Lecture Demonstrations: Fluid Dynamics for physical demonstrations.
- NPTEL: Fundamentals of Fluid Mechanics for Chemical and Biomedical Engineers for biomedical examples.
- University of Colorado Boulder Flow Visualization for experimental and artistic flow imagery.

APPENDIX: CARTESIAN VECTOR OPERATORS

For $\boldsymbol{u} = (u, v, w)$ and a scalar ϕ,

$$\nabla\phi = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right),$$

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z},$$

$$\nabla \times \boldsymbol{u} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}, \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}, \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right),$$

$$\nabla^2\boldsymbol{u} = \left(\nabla^2 u, \nabla^2 v, \nabla^2 w\right).$$

The vorticity is $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$.

APPENDIX: STRESS-DIVERGENCE IDENTITIES

With the traction convention $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$,

$$(\nabla \cdot \boldsymbol{\sigma})_i = \frac{\partial \sigma_{ij}}{\partial x_j}, \qquad \oint_{\partial\Omega} \boldsymbol{\sigma}\boldsymbol{n}\, dS = \int_\Omega \nabla \cdot \boldsymbol{\sigma}\, dV.$$

For constant μ,

$$\nabla \cdot (2\mu \mathbf{D}) = \mu \nabla^2 \boldsymbol{u} + \mu \nabla(\nabla \cdot \boldsymbol{u}).$$

For the full compressible Newtonian law with constant λ and μ,

$$\nabla \cdot \boldsymbol{\tau} = \mu \nabla^2 \boldsymbol{u} + (\lambda + \mu)\nabla(\nabla \cdot \boldsymbol{u}).$$

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- P. K. Kundu, I. M. Cohen, and D. R. Dowling, *Fluid Mechanics*, 6th ed. Academic Press, 2016. A standard alternative treatment of the same conservation-law derivation (further reading).

> Figure credit: `Stress.png` reproduces D. J. Acheson, *Elementary Fluid Dynamics* (1990), Figs. 6.2–6.3, for teaching use. All other diagrams and equations on these slides were prepared for these notes.